

October 31, 2024

Job Application for Software Engineer

Dear Recruitment Team,

I am a software engineer currently working at Samsung in low level systems. I started my career working in an embedded security startup customizing VeraCrypt bootloaders for ATM machines and modifying the BIOS to support remote management. After that, I started working at Samsung as a security analyst, where I would take part on security assessments of Android system services, drivers and applications for Samsung flagship devices. Two years ago I transitioned back into engineering by changing teams inside Samsung, and since then I have been working on low level systems, such as Trusted Applications running in ARM's TrustZone, Samsung's Android bootloader and providing consultancy in many of our projects.

Over the past 5 years I have acquired a lot of experience and made many contributions to projects I worked on that are a testament to my skills, all of which I am confident are a great fit for this position. Some highlights are:

- I implemented a macro-based testing framework for Trusted Application development in C, which helped us ensure the quality of our TA code;
- I reversed engineered the gRPC protocol used by Samsung's Bixby and implemented tools to facilitate eavesdropping communication between device and server to assist our security assessment of Bixby's systems;
- I led a major refactoring in Samsung Knox's Universal Credential Manager codebase in order to facilitate future extension;
- I worked in customizations in bootloaders for x86 and ARM64 architectures, both focused in adding security features;

Canonical sits atop the Linux world. Your engineering teams create and maintain the technologies that power most of the world's servers and devices overcoming many interesting and difficult challenges in the process. Moreover, Canonical is a leader in the Open Source community, showing the world that not only it is a viable business model, but also a successful one. While I don't have many contributions to the community, I am passionate by what it stands for and its many achievements, and I can see no better place to become more active and give back to the community than at Canonical.

I'm thrilled for an opportunity to contribute to Open Source and to Ubuntu in general, and hope that we can talk soon!

Sincerely,

Felipe S. Oliveira